Sheet 42 Tue. 9 Feb. 2027 Reverse percentage change; proportion and percentage change

1. Warm-up · Flash Calculate $\frac{1}{1{,}25}$.

2. Core · Method After an increase of 25%, which decrease returns to the initial value?

3. Core + · Interpret Of 250 candidates, 150 are admitted; last year there were 200 candidates. the proportion of applicants admitted and then the rate of change in the number of candidates.

4. Back · Table Table of values of f :

x	0	1	2	3
$f(x)$	4	1	0	1

Image of 2 ? Preimage of 1 ?

5. Running thread · MCQ On the globe, two points on the same parallel have:

A. the same latitude **B.** the same longitude **C.** the same distance at Greenwich **D.** the same altitude

Variations and extrema of a function

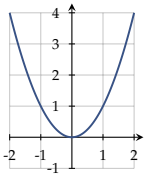
Sheet 43 Thu. 11 Feb. 2027 Increasing, decreasing, monotonic

1. Warm-up · Flash $f(x) = -2x + 5$. Calculate $f(1)$ and $f(3)$; what do we notice?

2. Core · MCQ f is increasing on I means that, for all $a < b$ in I :

A. $a \leqslant b$ only **B.** $f(a) \geqslant f(b)$ **C.** $f(a) \leqslant f(b)$ **D.** $f(a) = f(b)$

3. Core + · Graph Curve of $f(x) = x^2$ on $[-2;2]$:



Give the variation of f.

4. Back · Flash What multiplier corresponds to a 15% increase? To a decrease of 15 % ?

5. Running thread · Flash 457 divisible by 3? By 9? Justify.

Sheet 44 Mon. 1 st March 2027 Table of variations ; maximum and minimum over an interval

1. Warm-up · Flash Calculate x^2 for $x = -2, 0, 3$.

2. Core · Method f is decreasing on $[-3;1]$ with $f(-3) = 5$, $f(1) = -1$, then increasing on $[1;4]$ with $f(4) = 4$. Give the maximum and minimum of f on $[-3;4]$.

3. Core + · True/false True or false? Justify. The maximum of a function on $[a;b]$ is always attained at a or in b.

4. Back · Table Is this a table of proportional values?

x	1	2	3
y	5	9	13

5. Running thread · Method Is the triangle with sides 5, 12 and 13 right-angled? Which property should be used?